

Fiber Components of the Recurrent Laryngeal Nerve of the Rat: A Study by Light and Electron Microscopy

ÅKE DAHLQVIST, BENGT CARLSÖÖ, AND STEN HELLSTRÖM

Departments of Otolaryngology (Å.D., B.C.), Histology (Å.D., B.C.), and Anatomy (S.H.), University of Umeå, Umeå, Sweden

ABSTRACT A quantitative study by light and electron microscopy of the recurrent laryngeal nerve (RLN) of the rat has been performed. The diameter of the myelinated fibers displayed a bimodal pattern, with a pronounced peak at 2–3 μm , and with a second peak at 4.5–7 μm . The number of small fibers was reduced at the termination of the nerve, indicating that many of the small fibers are distributed to the trachea and oesophagus as the nerve ascends in the neck. Upon comparison, no difference between right and left side was noted in numbers of myelinated fibers. The RLN was also found to contain numerous unmyelinated fibers. Most of these fibers were of the diameter size range 0.2–0.6 μm . The unmyelinated fibers constituted about 50–60% of all axons in the RLN.

Quantitative studies by light microscopy on the recurrent laryngeal nerve (RLN) with regard to fiber size and number have been previously performed in several mammalian species (e.g., Tomasch and Britton, 1955; Murray, 1957; Scheuer, 1964; Gacek and Lyon, 1976; Harrison, 1981) as well as in avian nerves (Abdalla and King, 1978; Schwaber and Cohen, 1978).

Whereas numerous light microscopical investigations of this nerve have been published, comparatively few ultrastructural studies have appeared. Electron microscopical studies on RLN in the cat, the pigeon, and in humans have documented that the nerve contains motor and sensory fiber components as well as unmyelinated fibers (Gacek and Lyon, 1976).

In two recent studies (McDonald and Blewett, 1981; Carlsöö et al., 1982) granulated cells, morphologically similar to glomus cells of the carotid body, have been described within the RLN of the rat. What specific sensory information this endoneural microglomus in the recurrent nerve provides is at present not known.

To gain more fundamental knowledge of the neuronal pathways to and/or from the above-mentioned recurrent laryngeal nerve microglomus, more thorough information of the fiber composition of the recurrent laryngeal nerve of the rat is required. In the present study a quantitative analysis as well as an ultrastructural study of the RLN of the rat was per-

formed, special interest being focused on its fiber composition.

MATERIALS AND METHODS

Adult Sprague-Dawley rats weighing 250–300 gm were anesthetized with an intraperitoneal injection of sodium-pentobarbital (30 mg/kg body weight). The rats were then perfused through the left heart ventricle for 10 minutes with ice-cold 2% glutaraldehyde solution in 0.1 M sodium cacodylate buffer (pH 7.40, 440 mosm, flow rate 30 ml/min). The fixative was preceded by a 10-ml physiological saline wash. After aldehyde fixation the neck region was dissected under an operating microscope to localize the recurrent laryngeal nerve (RLN). In each rat, both nerves were excised from their entry into the larynx to their origin from the vagus nerves. The latter are located behind the subclavian artery on the right and the aortic arch on the left side. The length of the nerves was immediately determined and the nerves were then cut into segments about 5 mm in length, numbered in order, and further processed for microscopy. After a buffer (0.2 M sodium cacodylate) rinse, the specimens were postfixed in 1% osmium tetroxide in the same buffer, dehydrated in graded alcohol solutions, and embedded in Epon. Semithin sections were cut from the segments close

Received April 9, 1982; accepted July 26, 1982.

to the larynx and near the origin from the vagus nerve, picked up on glass slides, stained with toluidine blue, and examined under a light microscope. Cross sections from these speci-

TABLE 1. Average length of the RLN in four mammalian species

Species	Right	Left	Difference	Reference
Human	32 cm	43 cm	24%	Krmpotić (1958)
Giraffe	1.7 m	2 m	15%	Harrison (1981)
Dog	23 cm	36 cm	36%	Atkins (1973)
Rat	2.3 cm	3.2 cm	28%	Current study

mens were then photographed in a Zeiss photomicroscope.

For the morphometric analysis the paper prints were enlarged to an overall magnification of $\times 1,200$. The number of myelinated fibers was counted and their diameters were measured in a Zeiss TGZ-3 particle size analyser. The area of the circle of light of the particle size analyser was altered to coincide with the area of the profile of the fiber. Heavily distorted fibers were counted but not measured. For the EM studies, thin sections (about 70 nm) were collected on naked copper grids and contrasted with uranyl acetate, followed by lead citrate. These sections were examined in a Philips EM-300 electron microscope. A mon-

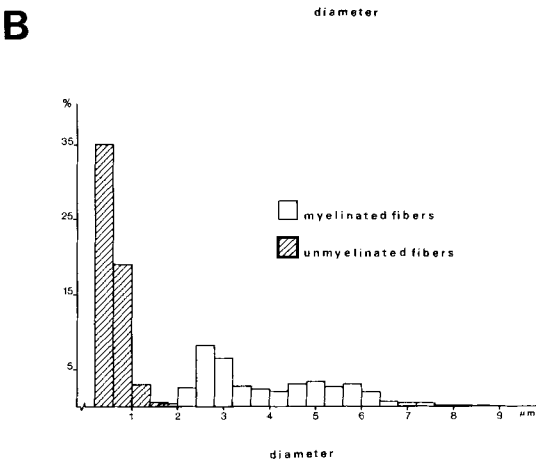
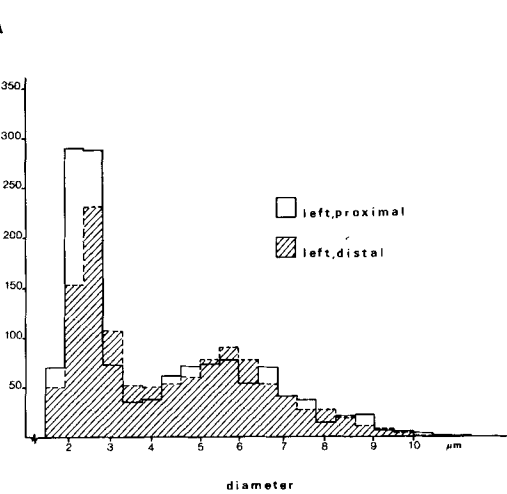
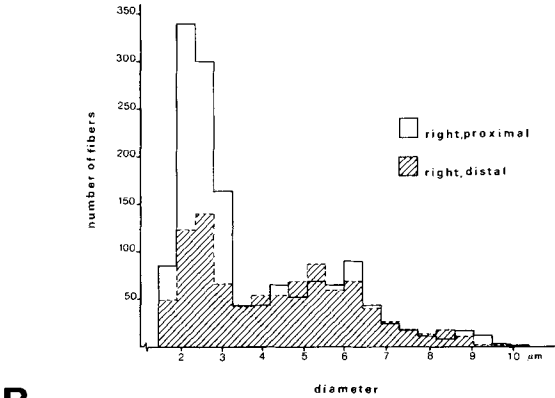
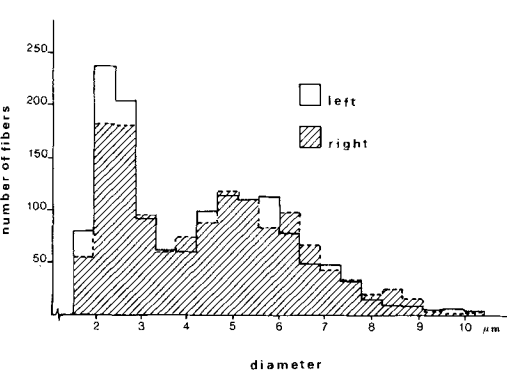


Fig. 1. Histograms of the diameter frequencies of myelinated fibers in A) distal portion of six right and left recurrent laryngeal nerves; B) four right recurrent laryngeal nerves, distal and proximal portion, respectively; C) four

left recurrent laryngeal nerves, distal and proximal portion, respectively; D) relative frequencies of myelinated and unmyelinated fibers in three recurrent laryngeal nerves.

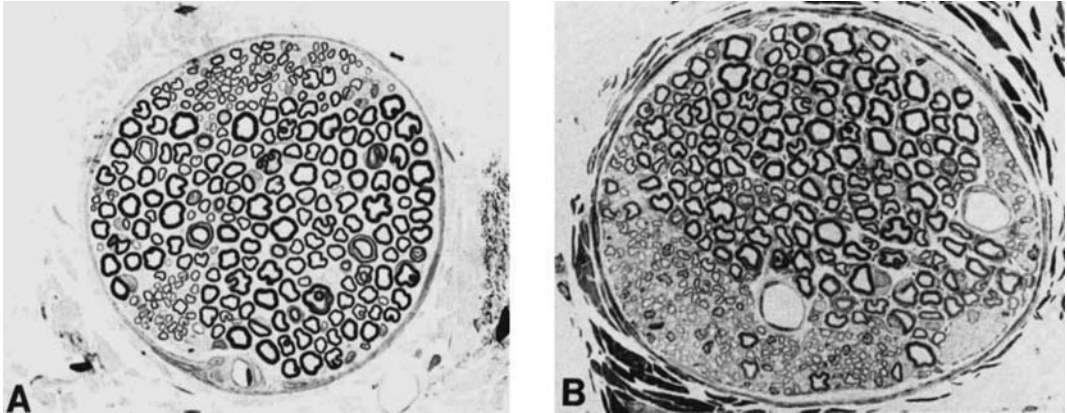


Fig. 2. Light micrograph of two different normal recurrent laryngeal nerves in the rat. A) distal portion, B) proximal portion. ($\times 450$)

tage of pictures for three nerves was arranged and by means of the particle analyser the number and individual diameters of all unmyelinated fibers were quantified.

RESULTS

The median length of the right RLN was found to be 2.3 cm (2.0–2.6) and that of the left, 3.2 cm (3.0–3.6); this amounts to an approximately 28% difference in length between the two sides. In Table 1 the lengths of the right and left RLN of four different mammalian species are listed.

As depicted in Figure 1A, fairly small differences were found between the right and left nerve with respect to both the diameter of the myelinated fibers and to their number. However, there were fewer fibers in the distal portion of the nerve (close to the larynx) than in the proximal part (close to the vagus) on both sides (Fig. 1B,C). Furthermore, a wide variation in number of fibers was obvious between different nerves. The lowest fiber count (192) was found in a right nerve, whereas the highest (542) was in a left nerve.

In transverse sections of the nerves (Figs. 2, 3) it was generally found that the myelinated fibers of varying diameters were grouped in different compartments of the nerve. This distribution pattern of myelinated fibers is characterized by the axons exhibiting the largest diameters occupying the central major portion of the nerve, whereas the smaller fibers are often found intermingled with numerous unmyelinated fibers in one to three small, rather

well defined areas close to the perineurium (Fig. 4).

Calculation of the diameter of the myelinated nerve fibers revealed a tendency toward a bimodal size frequency distribution, with a sharp peak at about 2–3.0 μm . A second, less pronounced peak was also evident between 4.5 and 7.0 μm . No difference between the right and left side was detected in this respect. However, the first, sharp peak is markedly diminished in the distal portion of the nerve.

The numbers of unmyelinated fibers on EM photographs in three different nerves were found to be 285, 366, and 554 respectively. The numbers of corresponding myelinated axons were 289, 339, and 290. The diameters of the unmyelinated fibers varied between 0.2 and 1.8 μm . Most fibers (60%) were found in the range 0.2–0.6 μm (Fig. 1D).

DISCUSSION

In the rat, the RLN on the right side leaves the vagus dorsal to the subclavian artery, whereas on the left side it curves around the aortic arch. The terminal branches of the nerve enter the larynx dorsal to the cricothyroid joint. The left nerve is thus longer than the right, and in rats we found a difference of about 0.9 cm between the two sides. In humans this difference averages 11 cm (Shin and Rabuzzi, 1971) in dogs, 13 cm (Atkins, 1973), and in giraffes, 30 cm (Harrison, 1981).

In the dog (Shin and Rabuzzi, 1971) as well as in man (Tomasch and Britton, 1955; Krmpotić, 1958) it has been shown that the

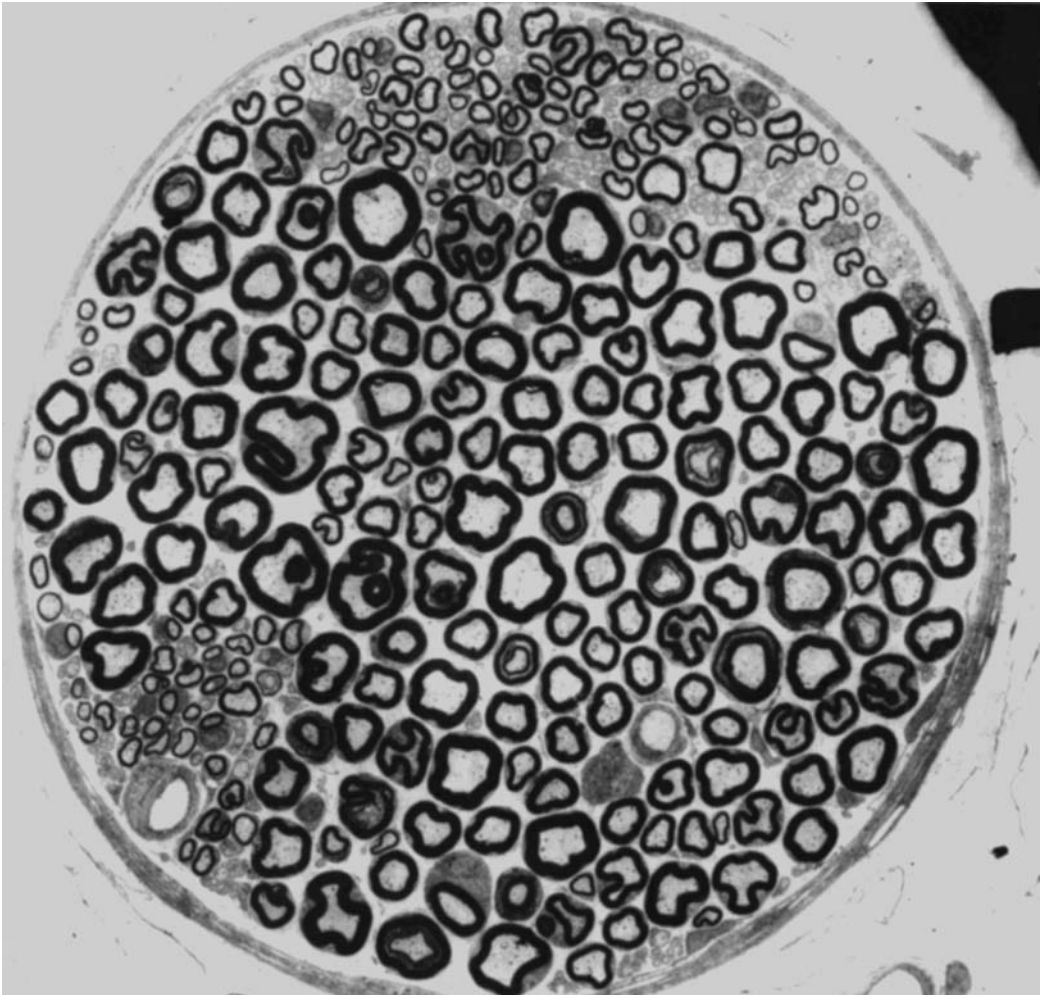


Fig. 3. Survey electron micrograph of a normal recurrent laryngeal nerve in the rat. Close to the perineurium, two areas occupied mainly by small myelinated fibers are seen.

The number of myelinated fibers was estimated to be 290. ($\times 1100$)

fiber diameters of the left RLN are greater than those of the right, allowing the simultaneous nerve impulses to travel faster on the left side and compensate for the difference in length between the two sides. In a conduction velocity study (Shin and Rabuzzi, 1957) a greater velocity on the left side was also found. However, Peytz et al. (1965) and Atkins (1973) found that conduction velocity is the same on both sides in the RLN. Furthermore, Atkins (1973) was able to demonstrate a slower velocity in the distal portion of the nerve compared to the proximal portion. Thus, the differences in con-

duction time between the two sides in the above-mentioned study may be explained by differences in the point of stimulation (Atkins, 1973). With regard to an average motor nerve conduction of 60–70 m/sec the difference in conduction time between the two sides in the rat could be calculated to 0.13 msec.

The cross-sectional area of the RLN as well as the number of fibers close to the vagus is greater than the same parameters close to the larynx. In the rat we found that mainly the myelinated fibers in the size range 2.0–3.0 μm were reduced in number in the more distal por-



Fig. 4. Electron micrograph. Numerous unmyelinated fibers are observed within the section. Myelinated fibers of

varying diameter can also be seen. Bar indicates $1\text{ }\mu\text{m}$ ($\times 13,000$)

tion. Chase and Ranson (1914) found in dogs that the RLN at its origin is composed of two sharply defined areas, one containing large, myelinated fibers and the other containing small, medium-sized myelinated fibers. Molhant (1910) demonstrated that the smaller fibers of the RLN were distributed in the cardiac, oesophageal, and tracheal branches and that the RLN as it enters the larynx is made up entirely of large myelinated fibers.

Similar observations have also been made in the cat (Murray, 1957). Murray also found that the branch of the RLN which enters the larynx has a unimodal fiber-size distribution with a fairly sharp peak at about $10\text{--}12\text{ }\mu\text{m}$ and relatively few fibers of smaller or larger diameter.

With regard to the number of fibers, it has been shown that the right nerve carries more myelinated fibers than the left. Abdalla and King (1979) demonstrated that the right nerve

of the hen contains 43% more than the left. In the feline RLN, too, Gacek and Lyon (1976) found a significantly larger number of fibers in the right nerve. The tendency for the right RLN to contain more fibers has also been documented by Abo-el-Enein (1967). However, in the present study no difference was observed between the right and left RLN with respect to number of fibers. Harrison (1981) found an equal number of myelinated nerve fibers in the right and the left RLN in man. Moreover, this author (Harrison, 1981) also counted the fibers in RLN of the giraffe and found no difference between the two sides. It is thus evident that there is no correlation whatsoever between bilateral fiber numbers and actual size of the animal.

Without the aid of the electron microscope an accurate count of unmyelinated fibers is impossible. Murray (1957), in a light microscopic study of the feline RLN, showed that the nerve contains relatively few nonmyelinated fibers and that the number in normal cats varied between 48 and 81 fibers. In a later ultrastructural study, Gacek and Lyon (1976) demonstrated that the actual unmyelinated fiber count varied (six animals) between 399 and 1256 in the feline RLN. Ravits et al. (1979) estimated the number of unmyelinated fibers per square millimeter at a transversely sectioned human RLN to 220,000. In the rat, no marked difference was found between the number of myelinated versus unmyelinated fibers in each individual nerve. We estimated the number of unmyelinated fibers to roughly 20,000 per square millimeter in the rat RLN. The origin of these nerves is unknown. However, it seems likely that they represent either cervical postganglionic sympathetic fibers entering the RLN in the neck, or postganglionic parasympathetic fibers whose cell bodies lie in the vagus nerve near the base of the skull. Further denervation studies might elucidate the origin of these unmyelinated fibers in the RLN.

ACKNOWLEDGMENTS

This investigation was supported by grants from the Swedish Medical Research Council (project number 19X-04002). The skillful tech-

nical assistance of Miss Siw Domeij is gratefully acknowledged.

LITERATURE CITED

- Abdalla, A.B., and A.S. King (1979) Afferent and efferent myelinated fibers in branches of the avian vagus. *J. Anat.*, 129:69-75.
- Abo-el-Enein (1974) Functional anatomy of the larynx. Ph.D. thesis. University of London. Quoted by R.E.M. Bowden. In: Ventilating and Phonatory Control Systems. B. Wyke, ed. Oxford University Press, London.
- Atkins, J.P. (1973) An electromyographic study of recurrent laryngeal nerve conduction and its clinical application. *Laryngoscope* 83:796-807.
- Carlsöö, B., S. Domeij, and S. Hellström (1982) Carotid body like tissue within the recurrent laryngeal nerve—an endoneurial chemosensitive microorgan? *Acta Otolaryngol. [Suppl] (Stockh)* 386:184-186.
- Chase, M.R., and S.W. Ranson (1914) The structure of the roots, trunk and branches of the vagus nerve. *J. Comp. Neurol.*, 24:31-60.
- Gacek, R.R., and M.J. Lyon (1976) Fiber components of the recurrent laryngeal nerve in the cat. *Ann. Otol. Rhinol. Laryngol.*, 85:460.
- Harrison, D.F.N. (1981) Fiber size frequency in the recurrent laryngeal nerves of man and giraffe. *Acta Otolaryngol. (Stockh)* 91:383-389.
- Krmpotić, J. (1958) Anatomisch-histologische und funktionelle verhältnisse des rechten und des linken Nervus recurrens mit Rückseit auf die Geschwindigkeit der Impulsleitung bei einer Ursprungsanomalie der rechten Schlüsselbeinarterie. *Arch. Ohr. Nasen u. Kehlk. Heilk.*, 1973:490-496.
- McDonald, D.M., and R.W. Blewett (1981) Location and size of carotid body-like organs (paraganglia) revealed in rats by the permeability of blood vessels to Evans blue dye. *J. Neurocytol.*, 10:607-643.
- Molhant, M. (1910) Le nerve vague. *Le Névrase*, 11:137.
- Murray, J.G. (1957) Innervation of the intrinsic muscles of the cat's larynx by the recurrent laryngeal nerve: An unimodal nerve. *J. Physiol.*, 135:206-212.
- Peytz, F., H. Rasmussen, and F. Buchtal (1965) Conduction time and velocity in human recurrent laryngeal nerve. *Dan. Med. Bull.*, 12:125-127.
- Ravits, J.B., A.E. Aronson, L.W. DeSanto, and P.J. Dyck (1979) No morphometric abnormality of recurrent laryngeal nerve in spastic dysphonia. *Neurology (NY)*, 29:1376-1382.
- Scheuer, J.L. (1964) Fiber size frequency distribution in normal human laryngeal nerves. *J. Anat.*, 98:99-104.
- Schwaber, J.S., and D.H. Cohen (1978) Electrophysiological and electron microscopic analysis of the vagus nerve of the pigeon, with particular reference to the cardiac innervation. *Brain Res.*, 147:65-78.
- Shin, T., and D. Rabuzzi (1971) Conduction studies of the canine recurrent laryngeal nerve. *Laryngoscope* 81:586-596.
- Tomasch, J., and W.A. Britton (1955) A fiber-analysis of the recurrent laryngeal nerve supply in man. *Acta Anat.*, 23:386-398.